Operations Research 1: Deterministic Models
Lecture 01: Introduction to Operations Research

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Fall, 2025

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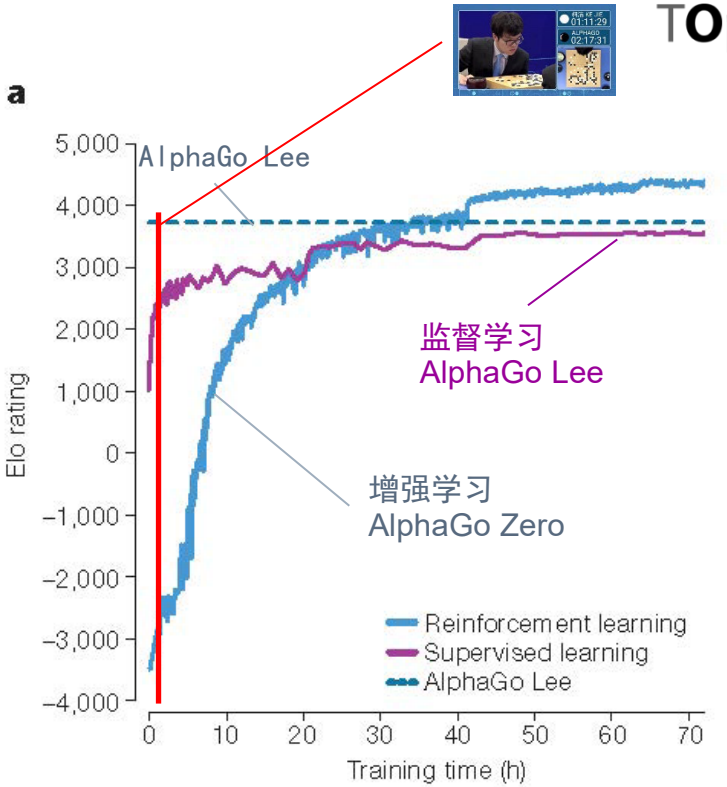
Tsinghua University, China

人工智能的一个重要里程碑：AlphaGo

- AlphaGo战绩
  - 2015.10
    AlphaGo Fan击败欧洲冠军樊麾（职业二段）5-0
  - 2016.03
    AlphaGo Lee击败世界冠军李世石（职业九段）4-1
  - 2017.05
    击败世界排名第一棋手柯洁（1997年出生，职业九段）3-0
  - 2017.10
    AlphaGo Zero击败AlphaGo **100-0**！
    3天训练：从零开始，“左右互博”！
    490万棋局 = （电脑）**19**局/秒 = （人类）**559**年
    （按照1局/小时，24小时/天计算）



ARTICLE

doi:10.1038/nature24270

Mastering the game of Go without human knowledge

David Silver[1]*, Julian Schrittwieser[1]*, Karen Simonyan[1]*, Ioannis Antonoglou[1], Aja Huang[1], Arthur Guez[1], Thomas Hubert[1], Lucas Baker[1], Matthew Lai[1], Adrian Bolton[1], Yutian Chen[1], Timothy Lillicrap[1], Fan Hui[1], Laurent Sifre[1], George van den Driessche[1], Thore Graepel[1] & Demis Hassabis[1]

Source: Silver et al. (2017)

运筹学与计算能力的快速发展

Robert Watson-Watt
(1892 – 1973)

Patrick Blackett
运筹学之父
(1897-1974)

1975年
CDC-7600
500万美元

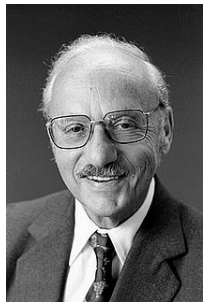
~3200万美元
（2013年）

成本降低：80,000倍

2013年
iPhone 4
400美元

1946
计算机
Penn

1980 1981 1984
PC PC lenovo
hp IBM DELL



George Dantzig
线性规划之父
(1914-2004)

1947
线形规划

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax \le b \\ & x \ge 0. \end{aligned}$$

单纯形算法

“二十世纪后半叶决策方法的革命性进步”
“二十世纪十大算法之一”

一战 二战

1900 1920 1940 1960 1970 1980 1990 2000 2010 2020

1908
工业工程

1937
运筹学

Source: Gass and Assad (2005, 2011)，Engineering and Technology History Wiki (ETHW)， McKinsey (2013)，Bixby (2017)等

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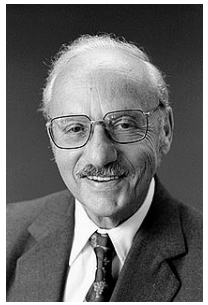

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单纯形算法
“二十世纪后半叶决策方法的革命性进步”
“二十世纪十大算法之一”

1988 – 2004年
计算提速： 1600倍
LP算法提速：3300倍
综合提速： 5,280,000倍
相当于：61天 ➔ 1秒

一战 二战

1900 1920 1940 1960 1970 1980 1990 2000 2010 2020

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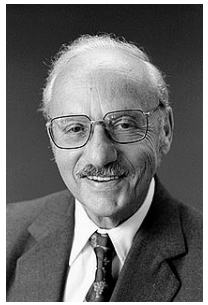

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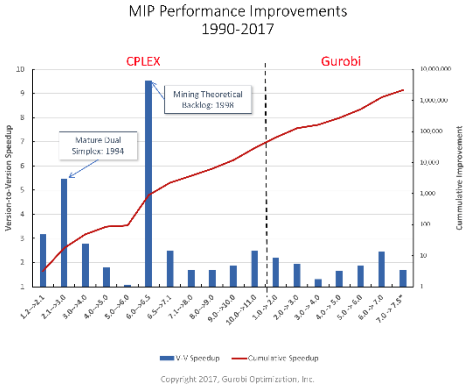

1990 – 2017年
MIP算法提速：1,000,000+倍

一战 二战

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1937
运筹学

Source: Gass and Assad (2005, 2011)，Engineering and Technology History Wiki (ETHW)， McKinsey (2013)，Bixby (2017)等

运筹学的起源
Origin of Operations Research (OR)

Origin of OR

- 雷达的军事应用（1936 – 1939）
  - 空军防务科学调研委员会（CSSAD, Tizard Committee）
  - 雷达技术：Radar（radio detection and ranging）
  - Bawdsey Manor on the Suffolk coast
  - Robert Watson-Watt
  - 1937年9月24日，世界上第一个雷达站
  - 提升拦截概率
    - 雷达：10倍
    - 运筹学分析：2倍
  - 成功要素：“人-机系统”
    - 科学家
    - 工程师
    - 皇家空军军官

Robert Watson-Watt[1]
(1892 – 1973)

Bawdsey Manor[2]

First towers (1937)[2]

Source: Gass and Assad (2011)，[1]Engineering and Technology History Wiki (ETHW)，[2]www.bawdseyradar.org.uk

Father of OR: Patrick Blackett

- 伦敦大轰炸（The London Blitz）
  - 打落一架敌机平均所需子弹数量
    - 1940年9月7日：20,000发
    - 1941年夏：4,000发
  - 实施步骤
    1. 基于雷达数据，预测敌机位置（纸、笔等）
    2. 简单的、可以快速生产的绘图机器
    3. 连接机械预测仪和雷达
    4. 调试机械预测仪
  - 成功要素
    - 多学科团队
      - 生理学家（3）、物理学家（1）、天体物理学家（1）、数学物理学家（2）、数学家（2）、测量员（1）、陆军军官（1）
    - 运筹学的应用

Patrick Blackett[1]
(1897 – 1974)

伦敦大轰炸（始于1940年9月7日）[2]

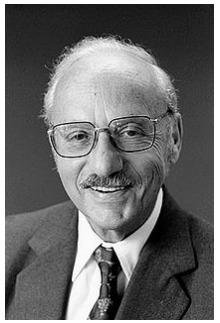
Source: [1]Gass and Assad (2011), [2]EyeWitness to History.com

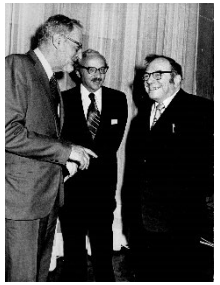
运筹学 = 源于实战 + 科学方法 + 跨学科团队合作

Operations research = real world practice + scientific methods
+ multi-disciplinary teamwork

Origin of OR

- Time zero: British military application of OR
  - 1936, British Air Ministry, Bawdsey Manor Research Station, Suffolk
  - 1937, as the origin of OR, set by the British OR Society in 1987
- Linear programming (LP)
  - 1939, production assignment problem as an LP, Leonid V. Kantorovich (Russian mathematician/economist)
  - 1947, general formulation with the simplex method, by George B. Dantzig
    - "Revolutionized decision making in the second half of the twentieth century"
    - "Top ten algorithms of the century"
- 1975 Nobel prize in economics
  - L. V. Kantorovich and T. C. Koopmans
  - The theory of optimum allocation of resources





Source: Gass and Assad (2011)

运筹学的定义
Definitions of OR

运筹学（Operations Research）的定义

- 为实施部门的运作决策提供定量分析的科学方法
  - A scientific method of providing executive departments with a quantitative basis for decisions regarding the operations under their control (Goodeve, 1948)

- 对政府、军事或商业运作中人力、机器、设备和政策的系统效率和性能表现的数学或科学分析
  - The mathematical or scientific analysis of the systematic efficiency and performance of manpower, machinery, equipment, and policies used in a governmental, military, or commercial operation (American Heritage Dictionary of the English Language, 1976)

- 决策科学
  - Science of decision making

Source: Gass and Assad (2011)

运筹学（Operations Research）与分析方法（Analytics）

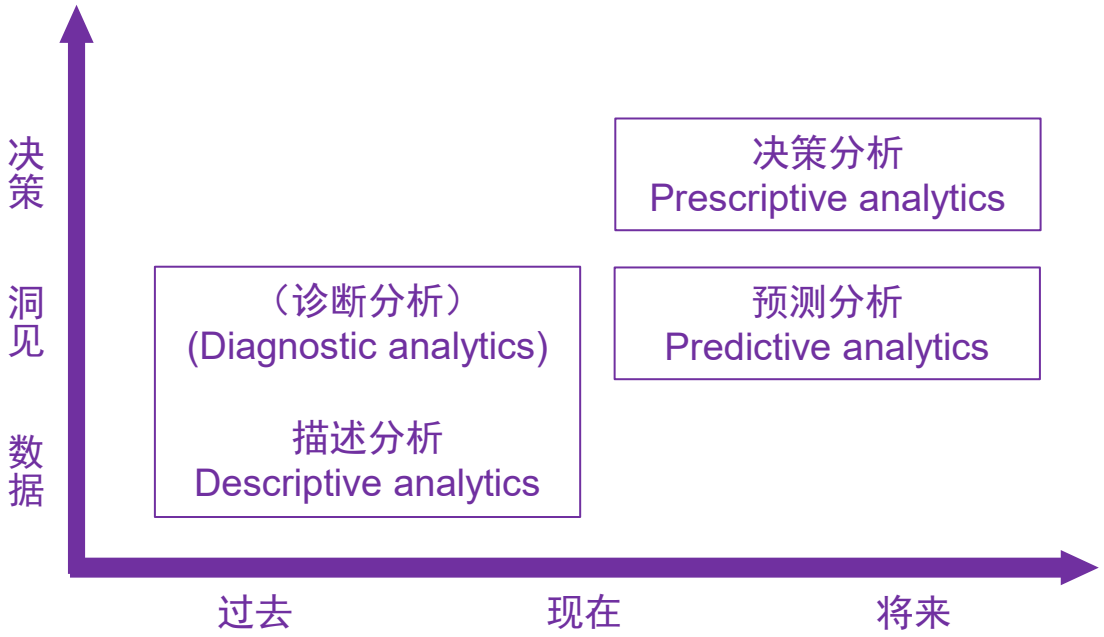
- 运筹学：将数据转化为洞察，并做出更好决策的科学过程
  - Operations research (OR) is defined as the scientific process of transforming data into insights to making better decisions.

- 分析方法：科学或数学方法在研究和分析复杂系统问题中的应用
  - Analytics is the application of scientific & mathematical methods to the study & analysis of problems involving complex systems.

Source: INFORMS

运筹学（Operations Research）与分析方法（Analytics）

- 描述分析 Descriptive analytics
  - 发生了什么？
- 诊断分析 Diagnostic analytics
  - 为什么会这样？
- 预测分析 Predictive analytics
  - 将来会怎样?
- 决策分析 Prescriptive analytics
  - 我们应该怎么做？



Source: IBM (2015), INFORMS

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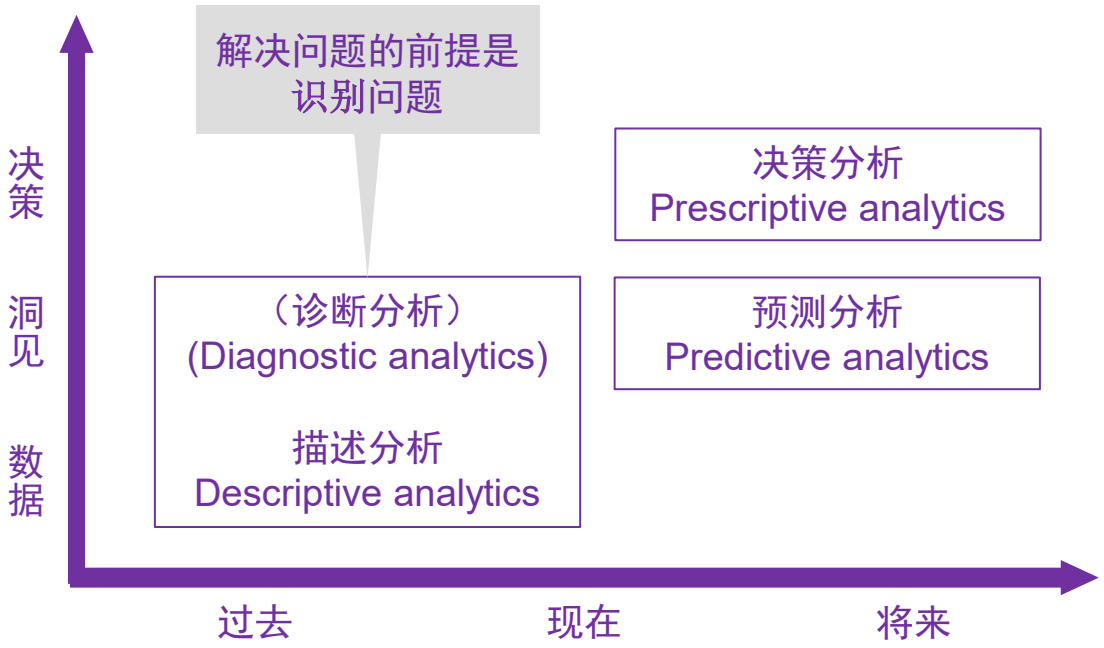

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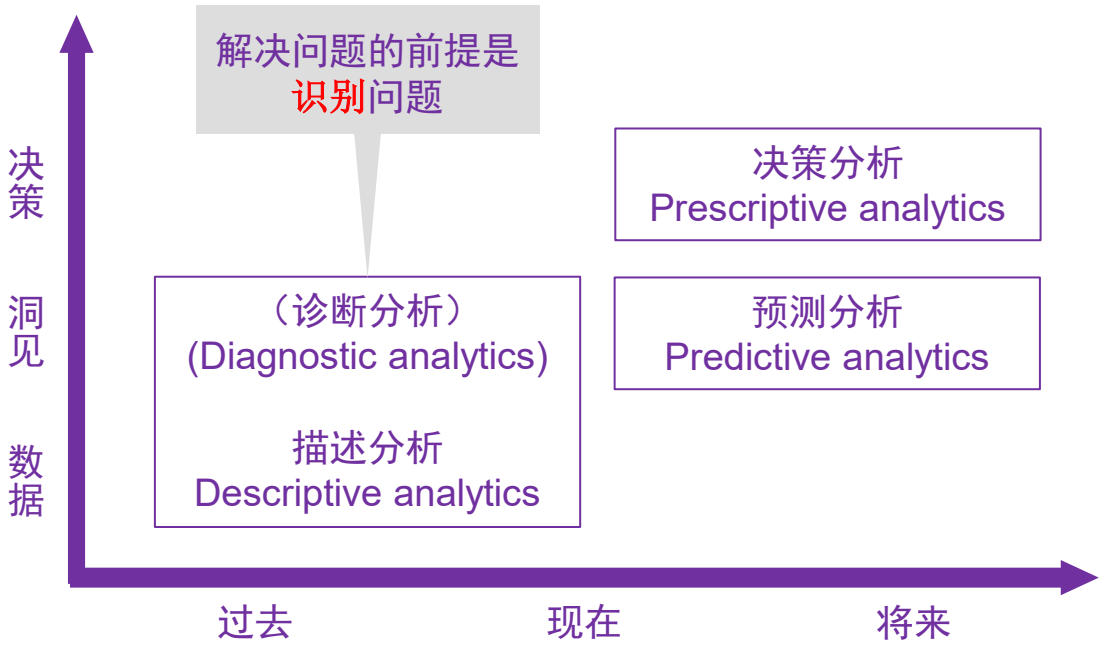

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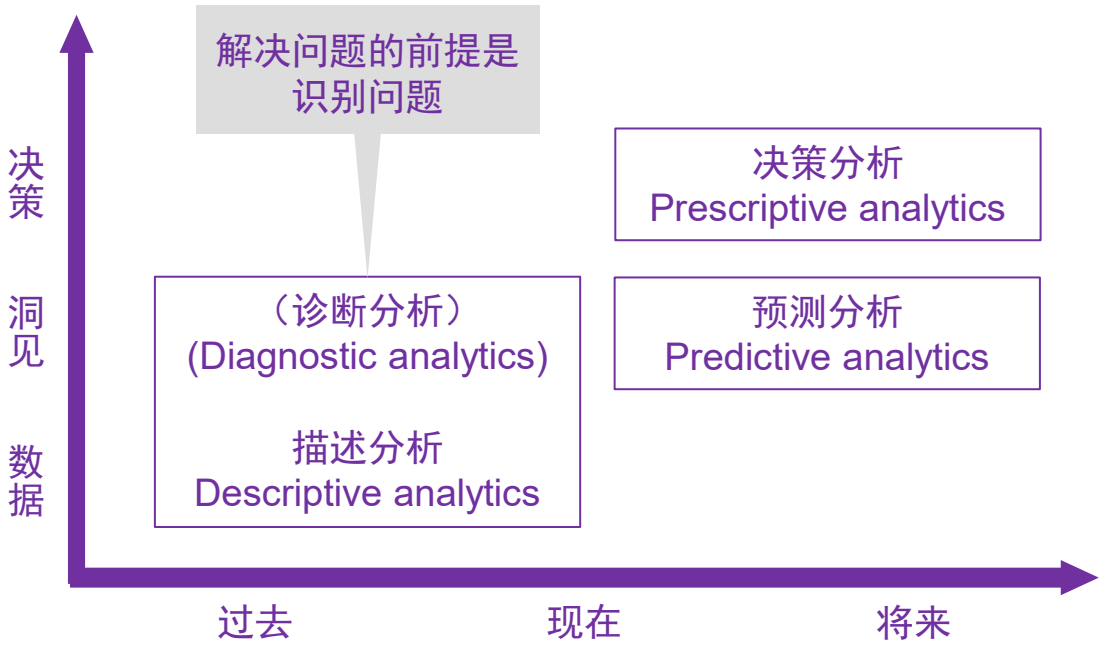


Source: IBM (2015), INFORMS

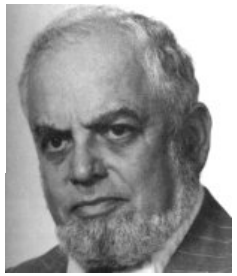
INFORMS Franz Edelman Award for Achievement in Advanced Analytics, Operations Research, and Management Science

Franz Edelman Award for Achievement in OR & MS

- Evaluation criteria
  - Implementation: implemented and in use?
  - Impact/Value: major quantified (e.g., dollars saved, revenue increased) and non-quantified (e.g., process streamlined, customer satisfaction improved) impacts? How important was the work to the client?
  - Technical Solution: technical innovation (creation of new methods or application of existing methods to new problems or new environments)?
  - Difficulty: political, technical, and managerial challenges
  - Transportability: portable to other applications or industries?

EDELMAN AWARD PAST WINNERS

2024 Molslinjen
2023 Walmart Global Tech
2022 Gobierno de Chile
2021 U.N. World Food Programme
2020 Intel Corporation
2019 Louisville Metropolitan Sewer District
2018 Federal Communications Commission
2017 Holiday Retirement
2016 UPS
2015 Syngenta
2014 U.S. Centers for Disease Control and Prevention
2013 Delta Programme Commissioner
2012 TNT Express
2011 MISO
2010 Indeval
2009 Hewlett-Packard
2008 Netherlands Railways
2007 Memorial Sloan Kettering Cancer Center
2006 Warner Robins Air Logistics Center
2005 General Motors
2004 Motorola, Inc.
2003 Canadian Pacific Railway
2002 Continental Airlines
2001 Merrill Lynch, Inc.
2000 Jeppesen Sanderson, Inc.
1999 IBM
1998 Bosques Arauco, S.A.
1997 Societé Nationale des Chemins de Fer Français and Sabre Decision Technologies
1996 South African National Defense Force
1995 Harris Corporation (Semiconductor Sector)
1994 Tata Iron & Steel Company, Ltd.
1993 AT&T
1992 New Haven Health Department
1991 American Airlines
1990 Health Care Financing Administration
1989 ABB Electric, Inc.
1988 City of San Francisco Police Department
1987 Syntex Laboratories, Inc.
1986 Southland Corporation (CITGO Petroleum Corporation Subsidiary)
1985 Weyerhaeuser Company
1984 Blue Bell, Inc. (dual)
1984 The Netherlands Rijkswaterstaat & the RAND Corporation (dual)
1983 Air Products and Chemicals, Inc.
1982 Arizona Department of Transportation
1981 ANR Freight System
1980 Kelly-Springfield Tire Company
1979 The Greater New York Blood Program
1978 Cahill May Roberts, Ltd.
1977 Syncrude Canada, Ltd.
1976 American Telephone & Telegraph
1975 Xerox Corporation
1974 Canadian National Energy Board
1973 The Babcock & Wilcox Company
1972 The Pillsbury Corporation



Source: https://www.informs.org/Recognizing-Excellence/INFORMS-Prizes/Franz-Edelman-Award

Franz Edelman Award for Achievement in OR & MS

- Manufacturing
  - Tata Steel Limited, India (2024): continuous steel annealing operation
  - General Motors (2022): vehicle content optimization
  - Lenovo (2021): production in large plants
  - Intel (2012): capital investment
  - P&G (2010): inventory optimization
  - *HP (2009): product portfolio management
  - *General Motors (2005): product throughput

Source: https://www.informs.org/Recognizing-Excellence/INFORMS-Prizes/Franz-Edelman-Award

Franz Edelman Award for Achievement in OR & MS

TOpS

- Healthcare
  - Janssen Pharmaceutical Companies of Johnson & Johnson (2022): data-driven COVID-19 Vaccine development
  - Merck Animal Health (2022): biomanufacturing productivities for life-saving medicine
  - Memorial Sloan Kettering (2021): radiotherapy cancer treatment planning
- Public Services
  - *Gobierno de Chile (2022): save lives during the COVID-19 crisis in Chile
  - U.S. Census Bureau (2022): reengineering field operations
  - Industrial and Commercial Bank of China (ICBC, 2011): branch reconfiguration
- Humanitarian
  - *The United Nations World Food Programme (2021): toward zero hunger

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Franz Edelman Award for Achievement in OR & MS

TOpS

- Transportation
  - Lufthansa Group (2025): airline schedule recovery
  - *Molslinjen, Danmark (2024): passenger ferry operations
  - American Airlines (2024): hub efficiency analytics tool (HEAT)
  - Danaos (2012): fleet-wide revenue routing
  - CSAV (2011): empty container logistics
  - *Netherlands Railways (2008): new timetable
  - *Canadian Pacific Railway (2003): operating plans
  - *Continental (2002): crew recovery

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Franz Edelman Award for Achievement in OR & MS

TOpS

- Supply Chain Management/Logistics
  - SF Express (2025): operations planning strategy
  - Amazon (2025): network design
  - McDonald's China (2024): supply chain network optimization
  - *Walmart (2023): outbound supply chain
  - DHL Supply Chain (2023): network mode, mixed fleet, and staging optimizations
  - JD.com (2023): socially responsible planning of SC capability
  - Alibaba (2022): integrated forecast, inventory, price optimization, & recommendation
  - Alibaba (2021): on-time hour-level delivery
  - JD.com (2021): unmanned warehouse robots
  - *UPS (2016): **ORION**
  - *TNT (2012): supply chain wide optimization (**GO**)
  - *DHL (2010): global brand investments
  - UPS (2003): next-day-air services network

Source: https://www.informs.org/Recognizing-Excellence/INFORMS-Prizes/Franz-Edelman-Award

Franz Edelman Award for Achievement in OR & MS

- Emerging applications
  - *USA Cycling (2025): USA cycling's women's team pursuit gold
  - Huawei Cloud (2023): saving network bandwidth cost for livestreaming services
  - Lyft (2023): matching of rider requests with drivers
  - Meituan (2023): real-time dispatching in food delivery

Source: https://www.informs.org/Recognizing-Excellence/INFORMS-Prizes/Franz-Edelman-Award

Franz Edelman Award for Achievement in OR & MS

- 2021
  - Alibaba
    - Alibaba VRP Algorithms Have Enabled Its On-time Hour-level Delivery
  - JD.com
    - Operations Research Optimization Algorithms Drive Unmanned Warehouse Robots to Work
  - Lenovo
    - Reinforcement Learning Greatly Improves Production Efficiency in Large Manufacturing Plants
  - Memorial Sloan Kettering, USA
    - Automated and Clinically Optimal Radiotherapy Cancer Treatment Planning
  - OCP
    - Global Food Security: Transforming OCP through Analytics
  - *The United Nations World Food Programme (WFP) – winner
    - Toward Zero Hunger with Analytics

Source: https://www.informs.org/Recognizing-Excellence/INFORMS-Prizes/Franz-Edelman-Award

Franz Edelman Award for Achievement in OR & MS

TOpS

- 2022
  - Alibaba
    - Integrated Forecast, Inventory, Price Optimization, and Recommendation Has Reduced Millions in Inventory and Shrinkage Costs for Alibaba Retail Businesses
  - General Motors
    - Vehicle Content Optimization at General Motors
  - *Gobierno de Chile – winner
    - Analytics Saves Lives During the COVID-19 Crisis in Chile
  - Janssen Pharmaceutical Companies of Johnson & Johnson
    - Data-driven COVID-19 Vaccine Development for Johnson & Johnson
  - Merck Animal Health
    - Operations Research Methods Transform Biomanufacturing Productivity for Life-saving Medicine
  - U.S. Census Bureau
    - Advanced Analytics Drives Reengineering of Field Operations for 2020 U.S. Census

Source: https://www.informs.org/Recognizing-Excellence/INFORMS-Prizes/Franz-Edelman-Award

Franz Edelman Award for Achievement in OR & MS

- 2023
  - DHL Supply Chain
    - Network Mode, Mixed Fleet, and Staging Optimizations for the DHL Supply Chain
  - Huawei Cloud
    - Huawei Cloud Adopts Operations Research for Livestreaming Services to Save the Network Bandwidth Cost:The GSCO System
  - JD.com, Inc.
    - Socially Responsible Planning Strengthens Supply Chain Capability for JD.com
  - Lyft, Inc.
    - A Better Match for Everyone: Reinforcement Learning at Lyft
  - Meituan
    - Meituan's Real-time Intelligent Dispatching Algorithms Build World's Largest Minute-level Delivery Network
  - *Walmart – winner
    - Optimizing Walmart's Outbound Supply Chain from Strategy to Execution – A Grocery Case Study

Source: https://www.informs.org/Recognizing-Excellence/INFORMS-Prizes/Franz-Edelman-Award

Franz Edelman Award for Achievement in OR & MS

- 2024
  - ALDI SUD Germany
    - Data-Driven Approaches and Collaborative Intelligence for Empowering Leadership of a European Market-Leading Retailer
  - American Airlines
    - HEAT – The Hub Efficiency Analytics Tool
  - McDonald's China
    - Supply Chain Network Optimization Has Reduced Millions in Transportation Costs and Fostered Economic Growth for McDonald's China
  - *Molslinjen – winner
    - Passenger Ferry Operations in the Digital Era: Forecasting and Revenue Management at Molslinjen
  - Tata Steel Limited
    - Optimization of Continuous Steel Annealing Operation Using Model Predictive Control at Tata Steel, India
  - Transvision
    - Optimizing Mobility for Elderly and Disabled Dutch Citizens Using Taxis

Source: https://www.informs.org/Recognizing-Excellence/INFORMS-Prizes/Franz-Edelman-Award

Franz Edelman Award for Achievement in OR & MS

- 2025
  - Amazon
    - Regionalize and Scale: Network Design for Faster and Cheaper Delivery
  - Flipkart, India
    - Delighting a Billion Indian Customers Through Supply Chain Optimization for E-commerce
  - Lufthansa Group
    - Decision Support for Airline Schedule Recovery
  - SF Express
    - SF Express Revolutionizes its Operations Planning Strategy Using Operations Research
  - *USA Cycling – winner
    - PROJECT 4:05 – Optimizing USA Cycling's Women's Team Pursuit Gold
  - WM
    - Dynamic Route Optimization and Automation of Industrial Routes at WM

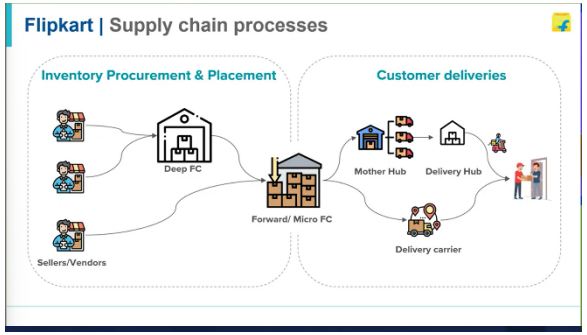
Source: https://www.informs.org/Recognizing-Excellence/INFORMS-Prizes/Franz-Edelman-Award

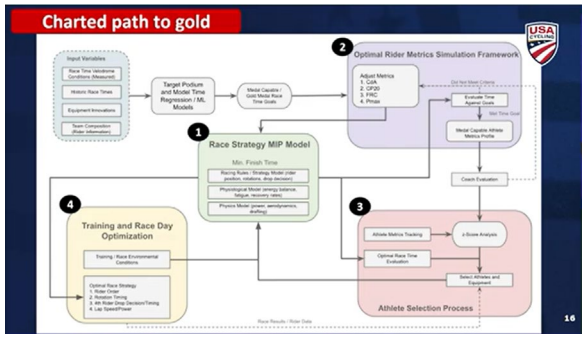
INFORMS Franz Edelman Award: Cases

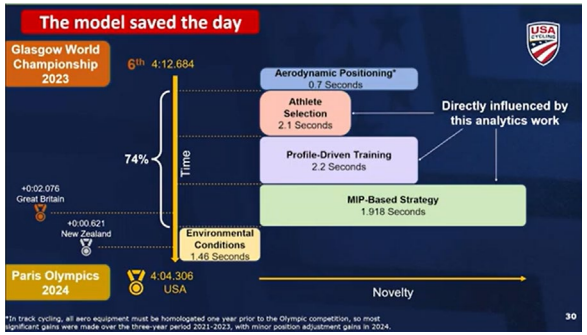
What are your favorite Olympic sports to watch?

USA Cycling (2025, Winner)

- Background (4’31”)
- Technical foundation (6’01”)
- Charted path to gold (1’55”)
- Results & impact (Project 4:05, 2’17”)
- MIP-based winning strategy
- Optimization in action: gold redemption (1’02”)
- The model saved the day (1’41”)
- Overcoming challenges



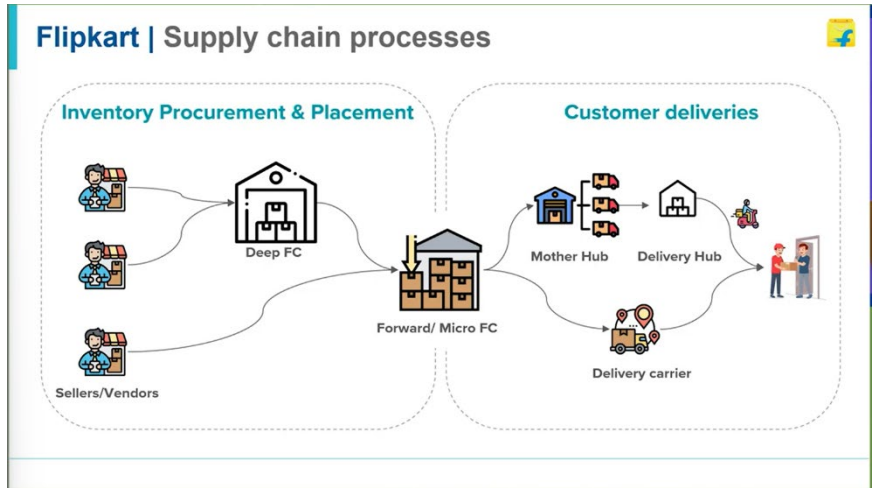


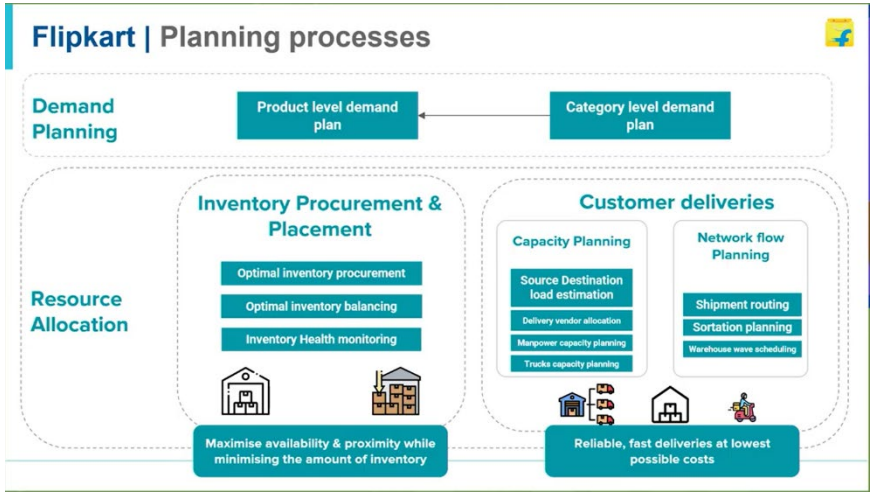


Source: https://www.informs.org/Recognizing-Excellence/INFORMS-Prizes/Franz-Edelman-Award

Flipkart (2025, Finalist)

- Company introduction (2’06”)
- Supply chain process (1’34”)
- Planning process (2’09”)

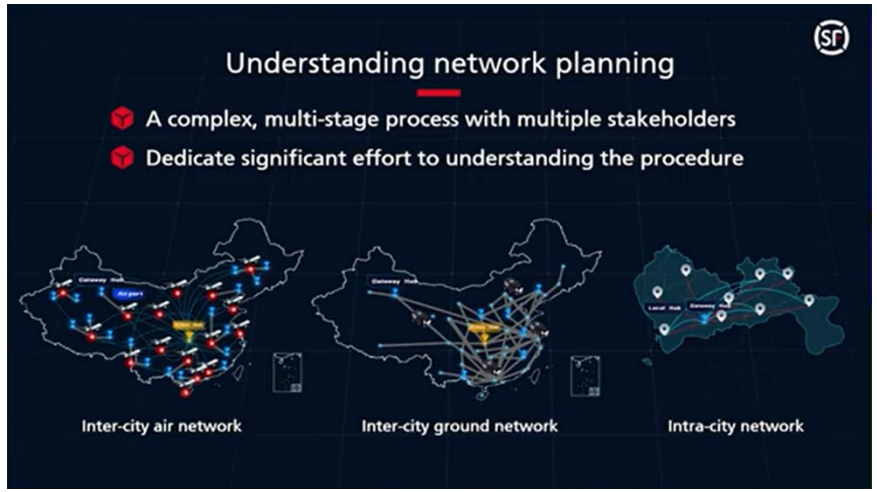


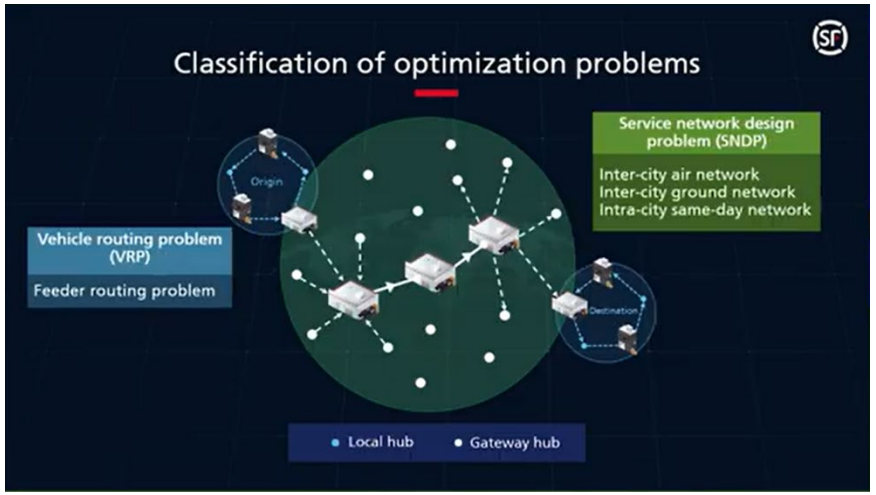


Source: https://www.informs.org/Recognizing-Excellence/INFORMS-Prizes/Franz-Edelman-Award

SF Express (2025, Finalist)

- Company background (2’03”)
- Network project (1’07”)
- SF network structure (3’56”)



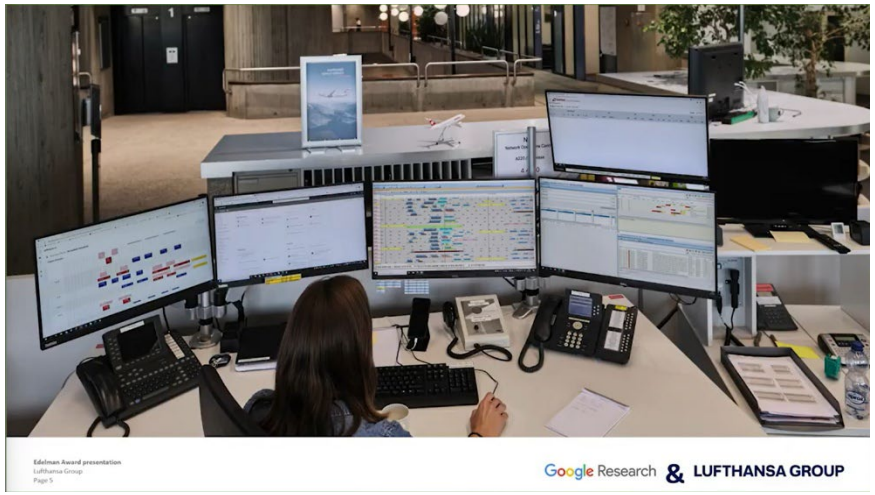


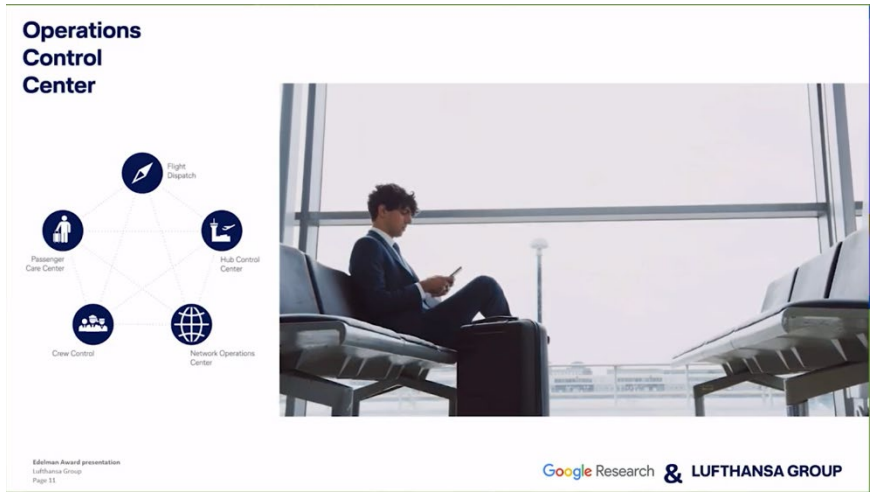
Source: https://www.informs.org/Recognizing-Excellence/INFORMS-Prizes/Franz-Edelman-Award

Have you ever experienced a flight delay or cancellation?

Lufthansa Group (2025, Finalist)

- Description of the disruption recovery process: stakeholders & considerations (3’40”)



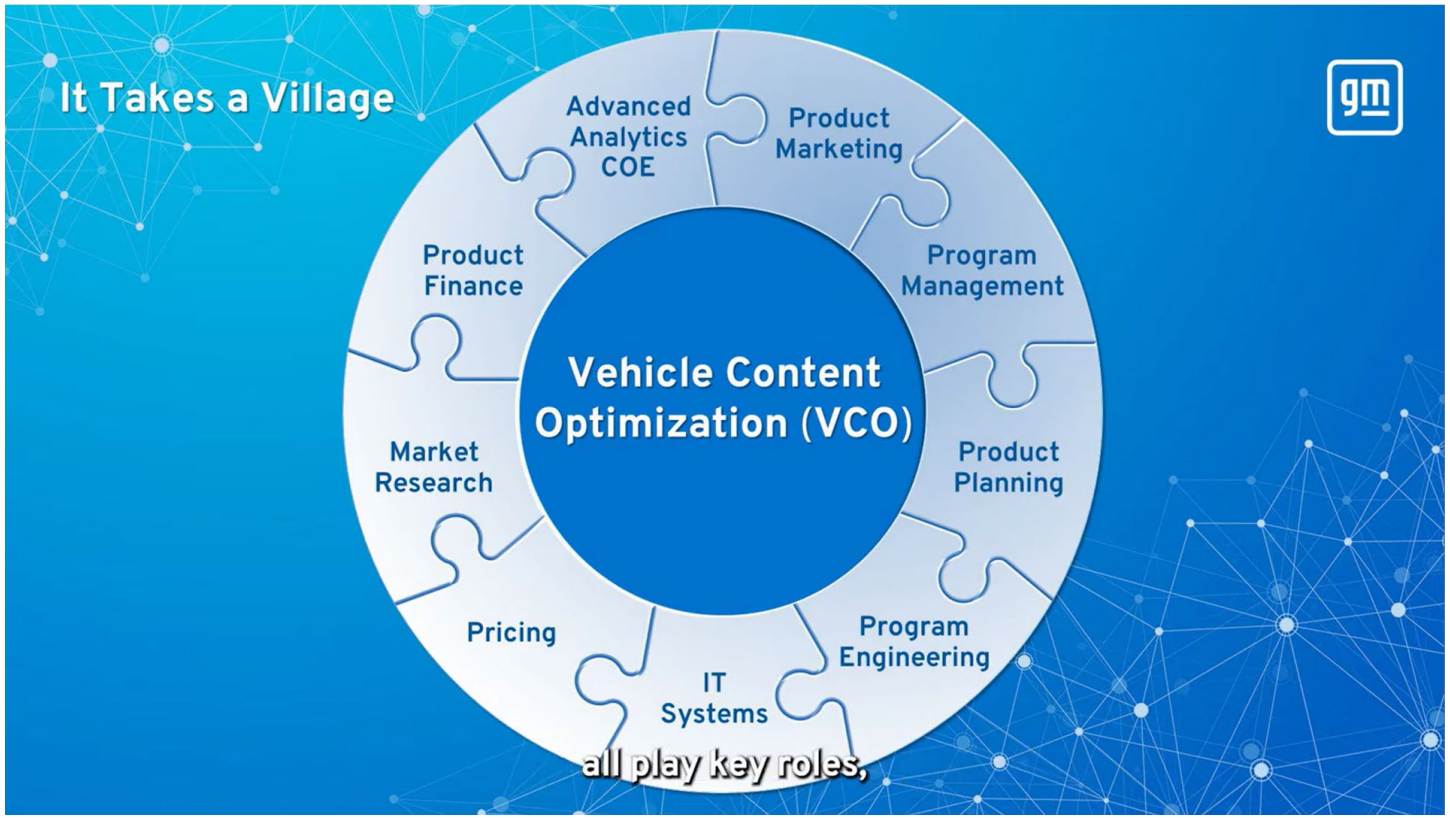


Source: https://www.informs.org/Recognizing-Excellence/INFORMS-Prizes/Franz-Edelman-Award

Just curious, what are your must-have features in a car?

General Motors (2022, Finalist)

- Vehicle content optimization (VTO, 3’40”)



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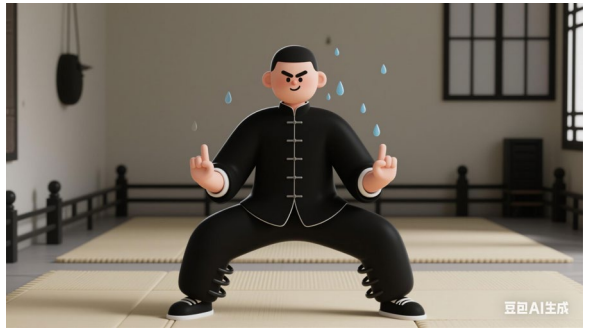
Operations Research Is Cool!



But it takes time and effort to learn and practice, starting from the basics.

Q & A